

GOVERNED KNOWLEDGE SYSTEMS · COMPANION SERIES · THESIS II

The Retrieval Covenant

A Contract Calculus for Warranted Recall between
Metabolic Memory and Time-Bounded Search

by

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Declaration

This manuscript is the second thesis of a companion series and, like its predecessor, a **design synthesis** presented to the conventions of a master's thesis. Its subject — the Retrieval Covenant — is a notional contract calculus: no measured implementation exists, and nothing here should be read as an empirical result. The two architectures it binds, the Metabolic Memory Architecture and the Symbiotic Search Fabric, are internal specifications of the same programme and are treated as given; where a constituent mechanism has an established basis in the published literature (axiomatic contracts, behavioural subtyping, anytime computation, working-set locality, freshness scheduling, provenance modelling), the text marks it *adopt, not invent* and cites the source. The claimed contribution is the covenant itself: one object, five clauses, an admissibility lattice, a warrant algebra, and a duality — the common chapter of memory and search, written once.

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Abstract

Two companion architectures now bracket the act of retrieval without owning it. On the memory side, the metabolic doctrine stratifies retention by residence time τ and admits recall as budgeted optimization — a datum is retrieved only when its expected contribution per unit cost clears a shadow price. On the search side, the Symbiotic Search Fabric stratifies access by deadline d and admits queries as contracts $\langle d, q_{\min}, b \rangle$ enforced by honest refusal. Each specifies retrieval from its own bank; neither gives it a spine of its own. The consequence is a composition gap: nothing guarantees that what memory undertakes to keep is what search is entitled to reach, that stratum validity predicates entail answer warrants, or that the two admission calculi price the same act consistently when one system's recall is the other's query.

This thesis extracts retrieval as a first-class discipline — the common chapter, written once. Its central object is the *retrieval covenant*: a bilateral contract joining a query's demand triple $\langle d, q_{\min}, b \rangle$ to a stratum's supply triple $\langle \tau, V, c \rangle$. Contributions: (i) the rung-stratum lattice, an admissibility structure establishing which deadline classes may bind to which residence tiers, with depth purchased rather than presumed; (ii) a warrant algebra deriving every answer's $\langle \text{TTL}, \text{confidence}, \text{provenance} \rangle$ stamp compositionally from the validity predicates of the strata that produced it — by meet, never by average; (iii) a duality schema — eviction is refusal issued in advance, storage is pre-paid retrieval, consolidation is caching under another name — allowing one shadow price to govern remembering and recalling as a single economy; and (iv) a closed-world binding registry and a lattice of six invariants integrating the covenant into both parent architectures without amendment to either. A four-phase validation programme with pre-registered falsifiers closes the work.

Keywords: retrieval contracts; residence time; deadline scheduling; warrant propagation; memory-search duality; design by contract; design science.

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Nomenclature

Symbol	Meaning
$D = \langle d, q_{\min}, b \rangle$	demand triple: deadline, quality floor, budget (the fabric's SP-1 contract)
$S = \langle \tau, V, c \rangle$	supply triple: residence class, validity predicate, cost profile of a stratum
$D \sqcap S$	a covenant: the bilateral binding of one demand to one or more supplies
$W = \langle \text{TTL}, \text{conf}, \text{cost}, \text{prov} \rangle$	warrant stamp carried by every answer
$\sqcap$	meet — conservative composition of warrants; never an average
$\oplus$	provenance concatenation (hash-linked)
$\langle d', q' \rangle$	counter-offer accompanying every refusal
$\lambda(t)$	the single shadow price rationing recall and retention alike
$E[q \mid d, b]$	expected answer quality under a demand, estimated from ledger history
R0–R3	deadline rungs: reflex ≤ 10 ms · interactive ≤ 1 s · deliberative ≤ 60 s · campaign ≤ 24 h
S0–S2	residence strata: hot/reflexive · working/tactical · archive/strategic
01–08	the eight covenant obligations (Chapter 3)
V1–V6	the six covenant invariants (Section 8.3)
RC-01...RC-16	edges of the closed-world binding registry (Table 8.1)
κ, ϵ	coupling consent and privacy budget, inherited from the fabric's membrane

1 · Introduction

Retrieval is the one act that memory and search both claim and neither owns. This chapter states the resulting problem, the research questions, and the contract-theoretic posture of the thesis.

1.1 Motivation

A programme now exists, across two prior specifications, for systems that manage what they know and how they come to know it: a metabolic memory doctrine that governs retention, eviction, and consolidation by residence time and a shadow price, and a search fabric that governs querying, coupling, and design evolution by deadline contracts and honest refusal. Between them sits a single verb — *retrieve* — used constantly by both and defined canonically by neither. Memory defines retrieval as the moment stored matter earns re-admission into action; search defines it as the moment a contracted question acquires a warranted answer. These are the same event described from opposite banks, and in any system that implements both doctrines the event happens once. A discipline that is specified twice will eventually be implemented twice, priced twice, and audited against two subtly different truths; the retrieval layer is where the programme would quietly fork. The motivation of this thesis is to prevent that fork by writing the common chapter once, as a contract.

1.2 Problem statement

Precisely: **specify a single, minimal object that both parent architectures can bind to without amendment, such that every retrieval in either doctrine is an instance of that object** — with its admission conditions, its service obligations, the derivation of its warrants, its refusal semantics, and its settlement duties stated once, closed-world, and falsifiably. The object must answer three questions the parents currently leave to goodwill: whether residence-time validity entails deadline-time warrant; whether what memory keeps is what search may reach; and whether the two admission calculi price the same act consistently.

1.3 Research questions

RQ1. What obligations does retrieval owe each parent, stated atomically and traceably? (Chapter 3.) **RQ2.** What single object discharges those obligations, and what admissibility structure governs which demands may bind to which supplies? (Chapters 4–5.) **RQ3.** How are answer warrants derived — with minimal ambiguity — from the validity of their sources, and what does the derivation imply about refusal? (Chapter 6.) **RQ4.** In what precise sense are remembering and recalling one economy, and how is the covenant bound into both parents and validated? (Chapters 7–9.)

1.4 Contributions

C1 — The composition gap, made explicit. Eight atomic obligations (O1–O8) that any shared retrieval discipline must discharge, with a traceability table (Chapter 3). **C2 — The covenant object.** A bilateral contract of five clauses joining demand $\langle d, q_min, b \rangle$ to supply $\langle \tau, V, c \rangle$, and the rung-stratum lattice that is its admission clause in tabular form (Chapters 4–5). **C3 — The warrant algebra and the duality.** A compositional derivation of every stamp by meet, minimum, and concatenation — never average — and a four-pair duality under which one shadow price lawfully governs both banks (Chapters 6–7). **C4 — Binding and validation.** A sixteen-edge closed-world registry integrating the covenant into both parents without amendment, six invariants, and a four-phase validation programme with pre-registered falsifiers (Chapters 8–10).

1.5 Methodology and evidentiary posture

The thesis follows the design-science paradigm [9, 20] and the contract lineage of computing: preconditions and postconditions as the meaning of an operation [11], contracts as the architecture of obligation [17], and substitutability as behavioural subtyping [14] — the covenant is, in that lineage, a behavioural type for the verb *retrieve*. Evidentiary posture is uniform: the whole work is Notional; adopted mechanisms are cited; the parents are internal specifications taken as fixed interfaces; and Chapter 10's falsifiers state what observations would break the design. As in the parent thesis, components precede relationships, and the binding registry of Chapter 8 is closed-world: an edge absent from it is prohibited.

1.6 Thesis outline

Chapter 2 reviews the five literatures the covenant draws on. Chapter 3 states the composition gap and the eight obligations. Chapter 4 specifies the covenant object; Chapter 5 its admissibility lattice; Chapter 6 the warrant algebra; Chapter 7 the memory–search duality. Chapter 8 binds the covenant into both parents and states the invariants. Chapter 9 gives the validation programme; Chapter 10 discusses falsifiability and limitations; Chapter 11 concludes and hands the price — assumed given throughout — to the third thesis of the series.

2 · Background and Related Work

Five literatures meet in the covenant: contracts, memory hierarchies, time-bounded computation, retrieval and its warrants, and the two parent architectures themselves.

2.1 Contracts in computing

The axiomatic tradition made an operation's meaning its precondition-postcondition pair [11]; design-by-contract made that pair an architectural instrument with named parties and blame [17]; behavioural subtyping made substitutability a provable relation between contracts [14]. The covenant imports all three: demand and supply are parties, the five clauses are pre/post structure, and the lattice of Chapter 5 is a substitutability statement — a stratum may serve a rung only if its supply contract behaviourally subsumes what the rung requires.

2.2 Memory hierarchies and locality

That storage is tiered and that programs exhibit locality are among the oldest quantitative facts of the field: the working-set model [5] and the engineered memory hierarchy [8] both teach that residence is a prediction about future use. The metabolic doctrine radicalizes the lesson — residence time becomes a governed, verified property with per-tier validity predicates — and the covenant consumes it as the supply side: a stratum is not a place but a promise $\langle \tau, V, c \rangle$.

2.3 Time-bounded computation

Anytime algorithms return the best answer available at interruption [4, 25]; hard-real-time scheduling admits work only against feasible deadlines [15]. The fabric's spine turned these into contract law; the covenant inherits that law as the demand side and adds what the spine left implicit — that the *supply* must also be contract-shaped, or the spine's guarantees rest on an unspecified party.

2.4 Retrieval and its warrants

Classical retrieval optimizes relevance over an index [1, 16]; agentic and retrieval-augmented systems consume answers as premises for further action [13, 24], which raises the stakes of an old question: what entitles a claim? Argumentation theory names the missing element the *warrant* — the licence connecting data to conclusion [21]. The covenant makes the warrant a first-class, machine-derived object: stamps $\langle \text{TTL}, \text{conf}, \text{prov} \rangle$ computed from source validity, with freshness decay in the tradition of synchronization scheduling [3], drift vigilance from the concept-drift and singular-learning literatures [7, 22, 12], provenance in the PROV lineage [18], and privacy debits under differential privacy when recall crosses tenants [6].

2.5 The two parents, and the gap

The Metabolic Memory Architecture (internal specification) governs retention with residence-time strata, per-tier validity, and shadow-priced admission; the Symbiotic Search Fabric (Thesis I of this series) governs querying with deadline rungs, honest refusal, and a closed-world interface registry. Each contains a retrieval fragment — the doctrine's optimization-triggered recall, the fabric's gate-and-descent — and the fragments presuppose each other without referencing each other. No published system known to this programme specifies retrieval as a bilateral contract between a retention regime and a deadline regime; multi-agent architectures [23] negotiate tasks, not warrants, and exit-capable federations [10] motivate but do not supply the mechanism. The gap is the covenant's to fill, and documented design rationale [19, 2] is the standard its exposition follows.

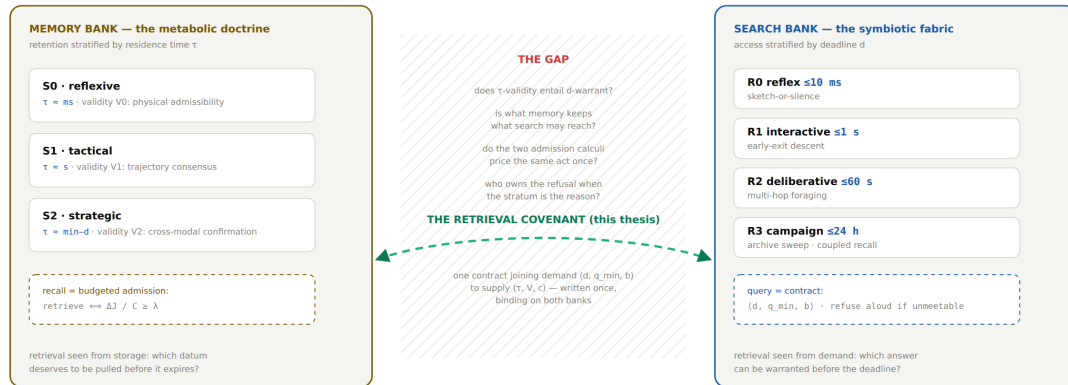
3 · The Composition Gap and the Eight Obligations

What exactly goes wrong if retrieval stays specified twice — and the atomic obligations any shared discipline must discharge.

3.1 The gap, stated operationally

Figure 3.1 draws the two banks. Three failures live in the hatched middle. **Entailment failure:** a datum valid by a stratum's predicate (physical admissibility at S0; consensus at S1; cross-modal confirmation at S2) is served into a contract whose floor $q_{\min}$ speaks a different language — nothing guarantees the translation. **Reachability failure:** memory evicts by its own calculus while search contracts assume the matter is still there; the fabric's refusal semantics have no way to say “memory declined to keep this,” so the refusal arrives late, unpriced, and unattributed. **Price failure:** the doctrine prices recall as admission-to-action, the fabric prices it as gate-passage; if the two λ diverge, the same act is cheap in one ledger and dear in the other, and arbitrage — real, systematic misallocation — follows.

The composition gap: retrieval specified twice, from opposite banks



each parent architecture specifies retrieval from its own side; the covenant is the common chapter extracted and given a spine

Figure 3.1: Retrieval specified twice, from opposite banks. Memory sees budgeted re-admission of stored matter; search sees contracted acquisition of warranted answers. The covenant is the bridge: one contract, binding on both.

3.2 The eight obligations

01 · Single specification. Retrieval is written once; both parents bind to the same text. *Design question:* what is the minimal object both can consume unchanged? **02 · Demand fidelity.** The demand triple is honored exactly: floor-or-refusal by the deadline, never silence. **03 · Supply fidelity.** A stratum serves only within its validity predicate V ; an answer a stratum cannot warrant is an answer it cannot give. **04 · Entailment.** Every binding carries a derivation from τ -validity to d -warrant; where no derivation exists, the binding is refused at admission — not discovered at delivery. **05 · Purchased depth.** Admissibility of deeper, slower supply is bought with deadline and budget; no rung presumes a stratum. **06 · Single price.** One shadow price values recall and retention; the two ledgers reconcile by construction. **07 · Warranted refusal.** A refusal is derived, priced, logged, and accompanied by the nearest honorable counter-offer $\langle d', q' \rangle$. **08 · Unconditional settlement.** Every outcome — serve, degrade, refuse — deposits a complete trace and books its cost; retrieval that leaves no residue did not happen, officially.

3.3 Traceability

Table 3.1: The eight obligations, their discharging constructs, and the invariants that bind them.

O	Obligation	Discharging construct	Bound by
01	single specification	the covenant object (Ch. 4)	closed-world registry (Table 8.1)
02	demand fidelity	clauses 2 & 4 (service, refusal)	V1, V4
03	supply fidelity	clause 3 + supply triple V	V1
04	entailment	warrant algebra (Ch. 6)	V1, V2
05	purchased depth	rung-stratum lattice (Ch. 5)	V3
06	single price	duality (Ch. 7) + λ hooks	V5
07	warranted refusal	refusal derivation (Sec. 6.4)	V4
08	settlement	clause 5 + RC-12...14	V6

4 · The Covenant Object

Two parties, two triples, five clauses, three lawful outcomes — and no fourth.

4.1 Parties

The **demand party** is any requesting agent — a forager executing a mission, a tuner probing regret, a distillery batch job — speaking the fabric's contract grammar. The **supply party** is the stratum steward: the curator authority responsible for a residence tier, speaking the doctrine's validity grammar. Neither party is human; both are agents under charters, and both sign. The covenant is bilateral by design: earlier drafts with a mediating broker reintroduced the very ambiguity the object exists to remove, because the broker accumulated discretion that belonged in the clauses.

4.2 The triples

Demand is the fabric's triple $D = \langle d, q_{\min}, b \rangle$ unchanged (O1 requires consuming parents' vocabularies, not paraphrasing them). Supply is the doctrine's stratum, contractualized: $S = \langle \tau, V, c \rangle$ — the residence class τ it maintains, the validity predicate V it can attest at serve time, and the cost profile c it will charge under the prevailing price. A stratum that cannot state its V has no standing to serve; this single requirement converts memory tiers from places into promises, and is the covenant's deepest demand on the memory side.

4.3 The five clauses

Clause 1 · Admission. A covenant forms only where the lattice permits the rung-stratum pair (Chapter 5) and the bid clears the price: $bid \geq \lambda(\tau)$. **Clause 2 · Service.** Execution is anytime: the answer in hand improves monotonically until d , and interruption at d is not an error but the contract's normal end [4, 25]. **Clause 3 · Warrant.** The stamp is derived from the serving strata's V by the algebra of Chapter 6 — meet, never average — and travels with the answer. **Clause 4 · Refusal.** If expected quality under the demand falls below the floor, $E[q \mid d, b] < q_{\min}$, the covenant refuses aloud with counter-offer $\langle d', q' \rangle$; degradation to the floor is lawful only when declared. **Clause 5 · Settlement.** Every outcome deposits the six-field trace into the ledger and books its cost; the deposit is unconditional (O8).

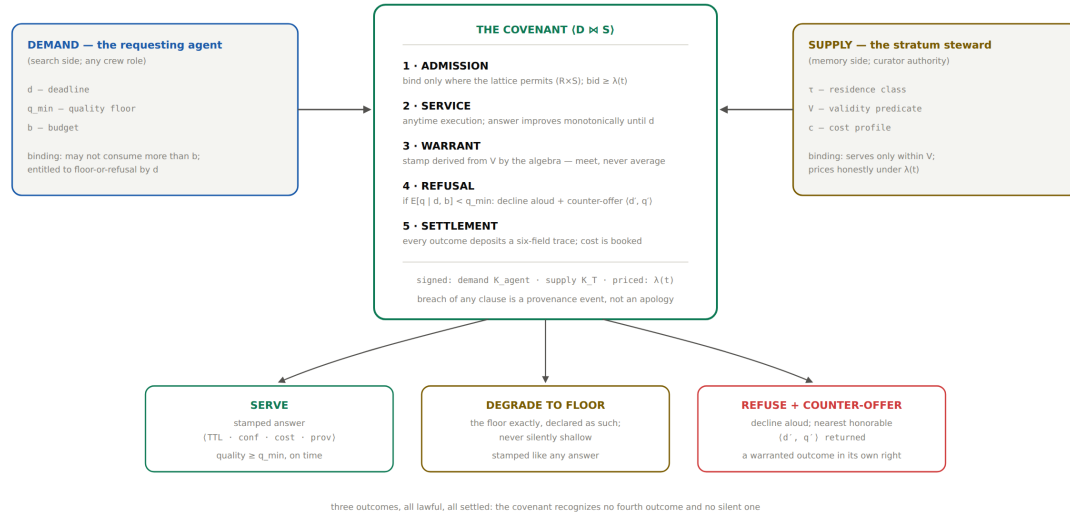
Anatomy of one covenant: two parties, five clauses, three lawful outcomes

Figure 4.1: Anatomy of one covenant. Demand and supply meet in five clauses; the three lawful outcomes are serve, declared degradation to the floor, and refusal with counter-offer. Breach is a provenance event, not an apology.

4.4 Breach semantics

A clause violated — a silent lateness, an unattested serve, a missing deposit — is recorded as a breach event on the provenance chain, attributed to the signing party, and priced against its charter's liability posture. The covenant does not attempt to make breaches impossible; it makes them *visible, attributable, and expensive*, which is what a contract is for [17]. Repeated breach patterns are exactly the distillate the parents' evolution machinery consumes; the covenant thus feeds the design loop it lives under without owning it.

5 · The Rung-Stratum Lattice

The admission clause in tabular form: which deadline classes may bind to which residence tiers, and why the forbidden cells are load-bearing.

5.1 The binding law

A demand at rung R may bind supply at stratum S only where the lattice marks the pair admissible. Admissibility is decided by one test with two prongs: *feasibility* — the stratum's cost profile admits an anytime schedule that fits inside d with non-degenerate expected quality; and *warrantability* — a derivation exists from V to a stamp the demand's floor can consume (O4). Both prongs are checkable at admission time from declared triples; neither requires trust.

The rung-stratum lattice: which deadlines may bind to which residence tiers

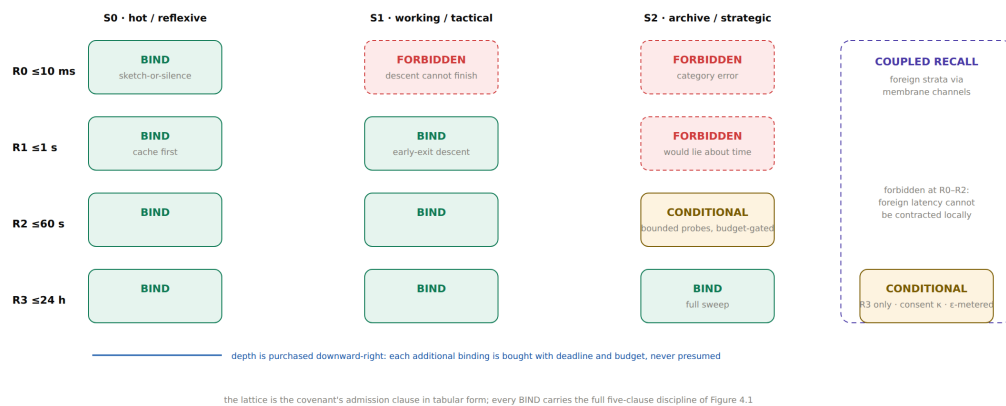


Figure 5.1: The lattice. Green cells bind; red cells are forbidden by feasibility or category; amber cells bind conditionally under explicit budget or consent. Coupled recall — foreign strata across the membrane — is admissible only at campaign timescale.

5.2 Reading the forbidden cells

The red cells carry the design's honesty. $R0 \times S1$ and $R0 \times S2$ are forbidden by feasibility: no admissible schedule completes a working-index descent, let alone an archive sweep, inside ten milliseconds, and a lattice that pretended otherwise would convert the reflex rung into a lottery. $R1 \times S2$ is forbidden for a subtler reason: a one-second budget *can* touch an archive, but only by sampling so thin that the resulting stamp would be indistinguishable from noise — the binding would be feasible and unwarrantable, and clause 3 must dominate. Forbidding a binding is not a denial of service; it is honesty about what the deadline can buy. The demand's remedy is the counter-offer: the refusal names the nearest rung at which the binding becomes lawful.

5.3 Conditional cells and coupled recall

$R2 \times S2$ binds under declared budget: bounded archive probes whose cost is escrowed at admission. Coupled recall — serving a demand from another tenant's strata — is a lattice column of its own, admissible only at $R3$ and only under the membrane's machinery: live consent κ , ε -metered exchange, incubated transit [6]. The rule beneath both: foreign latency and foreign validity cannot be contracted locally at interactive timescales, so the lattice refuses to let anyone promise them.

5.4 Monotonicity

Admissibility widens monotonically down the rungs: anything $R0$ may bind, $R1$ may bind, and so on. The lattice is therefore a genuine order structure — deeper time dominates — and “depth is purchased, never presumed” (O5) is its single generating principle. Substitutability runs the same direction: a stratum may stand in for a shallower one only if its supply contract subsumes the shallower promise [14], which in practice it never does for free — $S2$'s richer validity costs exactly the time $R0$ does not have.

6 · The Warrant Algebra

Every stamp is derived from the validity of its sources; the weakest source governs the whole.

6.1 The stamp grammar

Every answer carries $W = \langle \text{TTL}, \text{conf}, \text{cost}, \text{prov} \rangle$: a time-to-live, a confidence, the booked cost, and a provenance reference. The stamp is not metadata attached by courtesy; it is the covenant's clause-3 output, and an answer without a derivable stamp is not an answer (V1). The grammar names the warrant in exactly Toulmin's sense [21]: the licence that connects the served datum to the claim the requester will now act on.

6.2 Derivation rules

Three rules, one direction — conservative. **TTL by minimum:** the answer's TTL is the minimum of its sources' remaining validities, less transport and assembly time; freshness is a decaying asset and the clock starts at the oldest constituent [3]. **Confidence by meet:** constituent confidences compose by the meet $\sqcap$ — the intersection of what all sources jointly license — never by weighted average, because averaging destroys the worst-case bound that a floor comparison requires. When strata disagree, the claim narrows; it does not split the difference. **Provenance by concatenation:** hash-linked chains $\oplus$ -composed in serve order [18], so the stamp is auditable end-to-end by re-derivation.

The warrant algebra: every stamp is derived, compositionally, from the validity of its sources

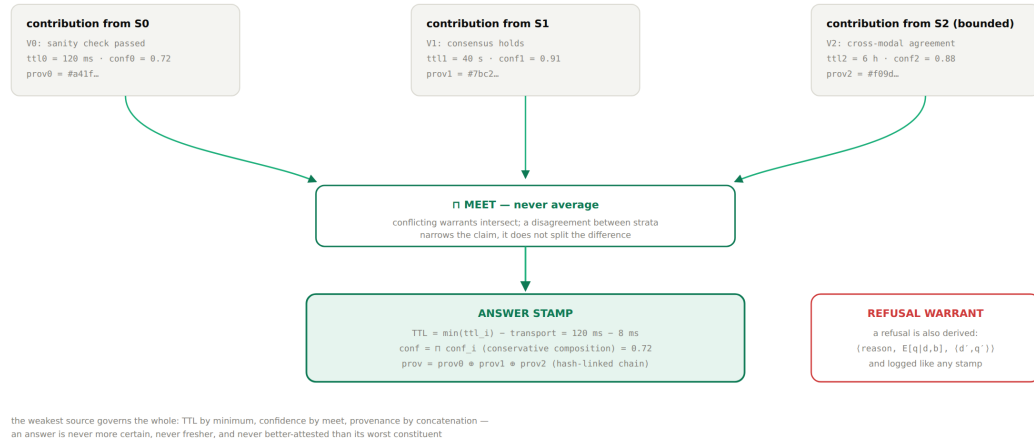


Figure 6.1: The algebra at work: three stratum contributions compose by meet into one stamp. TTL by minimum less transport; confidence by meet; provenance by concatenation. An answer is never fresher, surer, or better-attested than its worst constituent.

6.3 Why never average

The temptation to average is the temptation to manufacture warrant from disagreement. A floor test $\text{conf} \geq q_{\min}$ is a worst-case comparison; feeding it an average smuggles in an assumption of independence and good faith that no stratum attested. The meet rule is the algebra's one theorem-shaped commitment: *the composed warrant licenses exactly what every constituent licenses, and nothing more*. Its cost is honesty's usual cost — composed answers are weaker than their best part — and Chapter 10 registers the corresponding falsifier: if meet-composed stamps prove systematically, wastefully weaker than realized quality on replay, the algebra is too strict and must be weakened *by measured licence, not by default*.

6.4 The refusal warrant

Refusals are derived by the same machinery. A refusal carries $\langle \text{reason}, E[q \mid d, b], \langle d', q' \rangle \rangle$: the failed prong (feasibility or warrantability), the expectation that failed the floor, and the nearest honorable contract. It is stamped, logged, and settled like any serve (O7, O8) — a first-class outcome, because for the requesting agent a warranted “no” at ten milliseconds is often worth more than an unwarranted “yes” at a hundred. Drift vigilance guards the estimator itself: when the ledger-fitted $E[q \mid d, b]$ drifts [7], refusal rates move first, and the auditor treats a refusal-rate excursion as a drift alarm [22, 12].

7 · The Duality of Remembering and Recalling

Four verb pairs, one economy: why a single shadow price can lawfully govern both banks.

7.1 The four pairs

Eviction is refusal issued in advance. When a steward evicts, it declines — ahead of time and in bulk — every future demand that would have bound that matter; the honest version therefore posts a *pre-refusal notice* (RC-15) so the fabric's expectation estimator learns the supply has been withdrawn. **Storage is pre-paid retrieval.** To admit matter into a stratum is to buy, at today's price, a bundle of tomorrow's recalls; the working-set literature has always priced residence as predicted reuse [5, 8], and the covenant makes the prediction a booked position. **Consolidation is caching under another name.** Promoting matter to a stronger tier and seating an answer nearer its next use are the same act with different accents; the curator and the forager are trading in one market. **Residence τ and deadline d are two clocks on one lattice.** τ says how long matter stays actionable; d says how soon action must be answerable; the lattice of Chapter 5 is precisely their joint admissibility.

The duality: remembering and recalling are one economy read in two directions

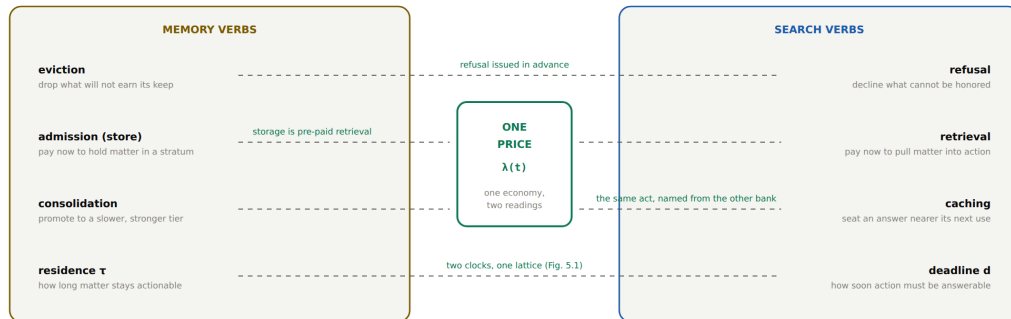


Figure 7.1: The duality: every memory verb is a search verb with the time axis reversed, and the covenant is the fixed point of the reversal. One price, two readings.

7.2 The one-price consequence

Because the four pairs identify the two banks' verbs, the two banks' prices must identify too — or the identification is broken somewhere measurable. This is the duality's operational content (O6, V5): the doctrine's admission test $\Delta J / C \geq \lambda$ and the fabric's gate test $bid \geq \lambda(t)$ draw on one ledger, and a persistent spread between what storage pays and what retrieval pays is arbitrage — evidence, not opinion, that the mapping mis-states one side. Chapter 9 makes the spread a monitored quantity and Chapter 10 makes it a falsifier. What the price *is*, where it comes from, and how it should move remain deliberately out of scope: the covenant consumes $\lambda(t)$ as given, and the series' third thesis owns its optimization.

7.3 Settlement symmetry

The duality extends to settlement: eviction deposits a trace exactly as retrieval does — what was dropped, under what price, against which pre-refusal notice — so the memory bank accumulates the same experience capital the search bank does. This symmetry is what lets the parents' shared evolution machinery reason about retention and access failures in one vocabulary; without it, half the system's regret would be invisible to the distillery.

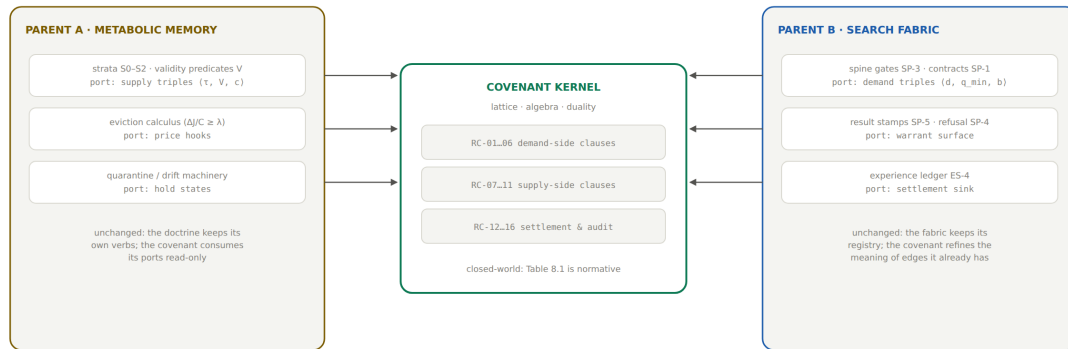
8 · Binding the Covenant into Both Parents

Integration by contract, not by surgery: the closed-world registry and the six invariants.

8.1 The binding principle

The covenant adds no component to either parent. It consumes the doctrine's supply ports (strata, validity predicates, eviction hooks) read-only; it refines the semantics of demand edges the fabric already has (SP-1 contracts, SP-4 refusals, SP-5 stamps); and it settles into ledgers that already exist (ES-4, SB-4). Integration is by contract, not by surgery — the parents keep their registries, and the covenant's registry is the sole place their vocabularies meet.

Binding the covenant into both parents — without amendment to either



the covenant adds no component to either parent: it consumes supply ports, refines demand semantics, and settles into ledgers that already exist — integration by contract, not by surgery

Figure 8.1: The covenant kernel between its parents. Ports are consumed, never amended; the sixteen RC edges of Table 8.1 are the complete and closed set of relationships.

8.2 The binding registry

Table 8.1: The covenant's closed-world registry. An edge absent from this table does not exist and must not be implemented.

ID	Edge	Payload	Cadence	Gate	Obl.
RC-01	agent → covenant	demand triple (d,q_min,b)	per query	signed; grammar-checked	O2
RC-02	covenant → lattice	rung-stratum admissibility query	per query	two-prong test	O5

ID	Edge	Payload	Cadence	Gate	Obl.
RC-03	λ source $\rightarrow$ covenant	prevailing price $\lambda(t)$	continuous	consumed as given	O6
RC-04	steward $\rightarrow$ covenant	supply triple (τ, V, c)	on change	signed pledge	O3
RC-05	covenant $\rightarrow$ both parties	admission seal	per binding	clauses 1-5 in force	O1
RC-06	kernels $\rightarrow$ agent	anytime answer stream	during service	monotone quality	O2
RC-07	steward $\rightarrow$ covenant	validity attestation at serve	per serve	V holds now	O3
RC-08	algebra $\rightarrow$ answer	warrant stamp W	per serve	derivation complete	O4
RC-09	covenant $\rightarrow$ agent	refusal + counter-offer (d', q')	on failed floor	derived; logged	O7
RC-10	covenant $\rightarrow$ agent	degradation declaration	on floor service	explicit marker	O2
RC-11	clock $\rightarrow$ stamp	TTL start at oldest constituent	per serve	min-rule	O4
RC-12	covenant $\rightarrow$ ledger	settlement trace (6 fields)	every outcome	unconditional	O8
RC-13	covenant $\rightarrow$ λ ledger	cost booking	every outcome	charged as sealed	O6, O8
RC-14	covenant $\rightarrow$ provenance	event append (seal, serve, breach)	always	hash-linked	O8
RC-15	steward $\rightarrow$ lattice	pre-refusal notice (eviction)	on eviction	duality duty	O7
RC-16	auditor $\rightarrow$ stamps	re-derivation sampling	continuous	independent check	O4

8.3 The six invariants

V1 · No unwarranted answer. Every served claim carries a stamp derivable, by RC-16 re-derivation, from attested validity. **V2 · Meet, never average.** Warrant composition is conservative; disagreement narrows claims. **V3 · Depth purchased, never presumed.** Only lattice-admissible bindings form, and admissibility widens monotonically with deadline. **V4 · Refusal is warranted.** Every refusal carries its derivation and a counter-offer; none is silent. **V5 · One price, both banks.** Retention and recall draw on a single λ ledger; persistent spread is a defect, not a market. **V6 · Settlement unconditional.** Every outcome deposits trace and books cost; unsettled retrieval is breach by definition. Each invariant names its enforcement in the registry (V1: RC-07/08/16; V2: RC-08; V3: RC-02; V4: RC-09/15; V5: RC-03/13; V6: RC-12/13/14).

9 · Validation Programme

Four phases, each with a burden that must be demonstrated — not asserted — before the next begins.

9.1 Phase α — Kernel and stamps (one estate, one stratum)

Steps. (1) Implement the covenant kernel against S1 only: triples, clauses, seals. (2) Implement the warrant algebra with RC-16 re-derivation audit from day one. (3) Wire settlement into an existing ledger. **Burden.** One hundred percent of served stamps re-derive identically under independent audit; no outcome escapes settlement under fault injection. **Exit.** A week of replayed workload with V1/V2/V6 continuously green.

9.2 Phase β — The full lattice and refusal

Steps. (1) Extend to S0 and S2; encode the lattice with its forbidden and conditional cells. (2) Bring refusal derivation live with counter-offers. (3) Adversarial demand mixes attempt forbidden bindings and floor-impossible contracts. **Burden.** Forbidden cells unreachable by construction (attempted binding fails closed); every refusal carries a well-formed $\langle d', q' \rangle$ that, when re-submitted, is admitted; no silent lateness under injected stalls. **Exit.** V3/V4 green across the adversarial suite.

9.3 Phase γ — The duality and the one price

Steps. (1) Hook the doctrine's eviction calculus and the fabric's gates to a single λ ledger. (2) Implement pre-refusal notices (RC-15) and eviction settlement. (3) Replay mixed retention/access workloads. **Burden.** Under the duality mapping, eviction decisions and refusal decisions coincide on matched cases; the store-versus-retrieve price spread stays within its declared band. **Exit.** V5 green; spread monitor armed as a permanent instrument.

9.4 Phase δ — Coupled recall at campaign timescale

Steps. (1) Open the R3 coupled-recall column under membrane consent, ε -metering, and incubation. (2) Extend stamps to carry foreign provenance segments. (3) Secession drill: verify a departing tenant's supply withdraws cleanly via pre-refusal notices. **Burden.** Foreign constituents never enter a stamp without incubation records; ε accounting exact under audit replay; post-secession, no stamp references withdrawn supply. **Exit.** The covenant operates federation-wide with all six invariants green — the steady state Thesis III's pricing work presupposes.

10 · Discussion

Answers to the research questions, four pre-registered falsifiers, and the honest limits of a notional contract.

10.1 Answering the research questions

RQ1 is answered by the eight obligations and their traceability (Chapter 3): retrieval's debts to each parent are atomic, named, and discharged by identifiable constructs. **RQ2** is answered by the covenant object and its lattice (Chapters 4–5): one bilateral contract, five clauses, admissibility as an order structure generated by “depth is purchased.” **RQ3** is answered by the warrant algebra (Chapter 6): stamps derive by minimum, meet, and concatenation, refusals derive by the same machinery, and ambiguity is confined to the one place it can be audited — the derivation itself. **RQ4** is answered by the duality and the binding registry (Chapters 7–8): remembering and recalling are one economy under one price, and the covenant enters both parents through sixteen closed-world edges without amending either.

10.2 Falsifiability

F1 · Entailment fails. If, in Phase α – β , a substantial class of bindings admits no derivation from τ -validity to d-warrant without new measurement, then O4 is unsatisfiable as designed and the covenant degenerates into per-pair adapters — the fork it was built to prevent. **F2 · The meet is too strict.** If meet-composed stamps are systematically and wastefully weaker than realized answer quality on replay, V2 is over-conservative and must be weakened by measured licence; if no measured licence exists, the algebra stands and the intuition falls. **F3 · The one price arbitrages.** If Phase γ 's spread monitor shows a persistent, exploitable spread between retention and recall pricing under the duality mapping, V5 is false in practice and the duality is decorative. **F4 · The lattice fights its users.** If forbidden cells attract sustained demand pressure ($R0 \times S2$ attempts dominating refusal logs), the rung boundaries — not the users — are mis-drawn, and the parents' evolution machinery should re-cut them. Each falsifier is measurable inside the Chapter 9 programme.

10.3 Limitations

The work is notional; both parents are internal specifications, so the covenant's fixed interfaces are stipulated rather than field-tested. Transport and assembly costs in the TTL rule are assumed measurable at stamp time — plausible on-estate, less so across the membrane. The estimator $E[q \mid d, b]$ carries all the usual risks of learned expectation under drift [7], mitigated but not removed by refusal-rate alarms. The lattice's rung and stratum boundaries are inherited, not derived; Chapter 10's F4 is the honest admission that they may be wrong. And the single-designer bias of the series applies here with extra force, since a contract's blind spots are precisely the cases its author did not imagine [19].

10.4 Hand-off to Thesis III

Throughout, $\lambda(t)$ has been consumed as given: clause 1 tests against it, clause 5 books against it, V5 reconciles across it. Where it comes from, what its components are, how it should move under load, what placement and reserves and federation do to it — all of that is deliberately absent here and constitutes the third thesis, *The Price of Recall*. The covenant fixes what retrieval *means*; the price fixes what it *costs*; the series closes when both are one system.

11 · Conclusion and Future Work

What was built, what it claims, and what should be done next.

11.1 Conclusion

Retrieval was the chapter both parent architectures wrote and neither owned. This thesis wrote it once: a bilateral covenant of five clauses joining the fabric's demand triple to the doctrine's contractualized supply; a lattice under which depth is purchased, never presumed; an algebra under which every stamp — and every refusal — is derived, conservatively, from attested validity; a duality under which eviction, storage, and consolidation are refusal, pre-payment, and caching read from the other bank, so that one price lawfully governs one economy; and a sixteen-edge closed-world registry that binds the whole into both parents without amending either. The object is small on purpose. A common chapter earns its place by being shorter than the misunderstandings it replaces.

11.2 Future work

Three lines follow directly. **Implement Phase α .** The kernel-and-stamps phase is a bounded engineering effort that would convert the algebra's re-derivation discipline from design into measurement. **Derive the lattice.** The rung and stratum boundaries are inherited constants; deriving them from workload statistics — and letting the parents' evolution machinery re-cut them under F4 pressure — would make the lattice a learned, governed object like everything else in the programme. **Price the covenant.** Thesis III should replace the assumed $\lambda(t)$ with a governed pricing mechanism and test V5's one-price claim under realistic, adversarial load; the covenant supplies the vocabulary and the falsifiers, and the series' arc — architecture, covenant, economy — completes there.

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Appendix A — Glossary

Table A.1: Terms of art used throughout the thesis.

Term	Definition
binding	the formation of a covenant between one demand and admissible supply (clause 1)
counter-offer $\langle d', q' \rangle$	the nearest honorable contract, returned with every refusal (clause 4)
covenant	the bilateral retrieval contract $D \bowtie S$: five clauses, three lawful outcomes
demand triple D	$\langle d, q_{\min}, b \rangle$ — deadline, quality floor, budget; the fabric's contract, consumed unchanged
entailment (O4)	the required derivation from a stratum's τ -validity to a demand's d-warrant
lattice	the rung-stratum admissibility structure; the admission clause in tabular form
meet Π	conservative warrant composition: what all sources jointly license, nothing more
pre-refusal notice	the eviction-time posting that withdraws supply from future bindings (RC-15)
purchased depth (O5)	the rule that deeper supply is bought with deadline and budget, never presumed
settlement	the unconditional deposit of trace and booking of cost for every outcome (clause 5)
stamp W	$\langle \text{TTL}, \text{conf}, \text{cost}, \text{prov} \rangle$ — the derived warrant carried by every answer
steward	the curator authority that signs a stratum's supply triple and attests V at serve time
supply triple S	$\langle \tau, V, c \rangle$ — residence class, validity predicate, cost profile; a stratum as a promise
warrant	in Toulmin's sense: the licence connecting served data to the claim acted upon
one-price rule (V5)	retention and recall draw on a single λ ledger; persistent spread is a defect